

CUSTOMER CHURN PREDICTION & LIFETIME VALUE (LTV) ENGINE

Machine Learning-Based Customer Retention Analytics Platform

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Internship: Data Analytics Internship

Tech Stack: Python | PostgreSQL | Scikit-learn | FastAPI | Metabase | SQLAlchemy | Git

Executive Summary

Customer retention is a critical challenge for telecommunication companies, where acquiring new customers is significantly more expensive than retaining existing ones. Businesses need predictive analytics solutions that identify customers at risk of leaving while also estimating the long-term value of each customer.

The **Customer Churn Prediction & Lifetime Value (LTV) Engine** is an end-to-end machine learning project designed to address these business needs. The system predicts whether a telecom customer is likely to churn and estimates the customer's future lifetime value (LTV). These predictions enable businesses to prioritize retention efforts, optimize marketing campaigns, and improve long-term profitability.

The project follows a complete data analytics workflow beginning with data collection and preprocessing, followed by feature engineering, machine learning model development, evaluation, deployment through FastAPI, storage of predictions in PostgreSQL, and visualization using Metabase dashboards. This demonstrates how data science can be integrated into real-world business applications to support strategic decision-making.

1. Project Overview

Introduction

Telecommunication companies generate vast amounts of customer data every day, including demographic details, service subscriptions, contract information, payment history, and billing records. Analyzing this information provides valuable insights into customer behavior.

Customer churn occurs when a customer decides to discontinue using a company's services. High churn rates reduce revenue and increase the cost of acquiring replacement customers. Predicting churn before it occurs allows businesses to take preventive actions such as personalized offers, improved customer support, or contract incentives.

Customer Lifetime Value (LTV) represents the total expected revenue a customer will generate during their relationship with the company. Estimating LTV helps businesses identify high-value customers and allocate retention resources effectively.

This project combines churn prediction and LTV estimation into a unified analytics platform, enabling organizations to make proactive, data-driven decisions.

2. Business Problem

Telecommunication providers face several challenges related to customer retention:

- Customers can easily switch to competing providers.
- Marketing budgets are often spent without identifying high-risk customers.
- Retention efforts may target low-value customers instead of high-value ones.
- Traditional reporting identifies churn only after customers have already left.

These challenges reduce revenue and customer satisfaction.

The objective of this project is to develop an intelligent analytics solution capable of predicting customer churn and estimating customer lifetime value before business losses occur.

3. Objectives

The primary objectives of this project are:

- Develop a machine learning model to predict customer churn.
- Estimate customer lifetime value (LTV).
- Store predictions in a PostgreSQL database.
- Provide REST APIs using FastAPI for real-time predictions.
- Build interactive dashboards in Metabase for business users.
- Support proactive customer retention strategies using predictive analytics.

4. Dataset Description

The project uses the **IBM Telco Customer Churn Dataset**, which contains customer demographic information, account details, subscribed services, and billing records.

Dataset Characteristics

Attribute	Description
Total Records	~7,043 customers
Target Variable	Churn (Yes/No)
Numeric Features	Tenure, MonthlyCharges, TotalCharges
Categorical Features	Gender, Contract, InternetService, PaymentMethod, etc.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U
1	customerI	gender	SeniorCitiz	Partner	Dependent	tenure	PhoneServ	MultipleLi	InternetSe	OnlineSeci	OnlineBac	DevicePro	TechSupp	Streaming	Streaming	Contract	PaperlessE	PaymentMethod	MonthlyCharge	TotalChar	Churn
2	7590-VHVI	Female	0	Yes	No	1	No	No phone	DSL	No	Yes	No	No	No	No	Month-to- Yes	Electronic check	29.85	29.85	No	
3	5575-GNV	Male	0	No	No	34	Yes	No	DSL	Yes	No	Yes	No	No	No	One year	No	Mailed check	56.95	1889.5	No
4	3668-QPYI	Male	0	No	No	2	Yes	No	DSL	Yes	Yes	No	No	No	No	Month-to- Yes	Mailed check	53.85	108.15	Yes	
5	7795-CFOI	Male	0	No	No	45	No	No phone	DSL	Yes	No	Yes	Yes	No	No	One year	No	Bank transfer (aut	42.3	1840.75	No
6	9237-HQIT	Female	0	No	No	2	Yes	No	Fiber optic	No	No	No	No	No	No	Month-to- Yes	Electronic check	70.7	151.65	Yes	
7	9305-CDSh	Female	0	No	No	8	Yes	Yes	Fiber optic	No	No	Yes	No	Yes	Yes	Month-to- Yes	Electronic check	99.65	820.5	Yes	
8	1452-KIOV	Male	0	No	Yes	22	Yes	Yes	Fiber optic	No	Yes	No	Yes	No	Yes	Month-to- Yes	Credit card (autorr	89.1	1949.4	No	
9	6713-OKO	Female	0	No	No	10	No	No phone	DSL	Yes	No	No	No	No	No	Month-to- No	Mailed check	29.75	301.9	No	
10	7892-POO	Female	0	Yes	No	28	Yes	Yes	Fiber optic	No	No	Yes	Yes	Yes	Yes	Month-to- Yes	Electronic check	104.8	3046.05	Yes	
11	6388-TABI	Male	0	No	Yes	62	Yes	No	DSL	Yes	Yes	No	No	No	No	One year	No	Bank transfer (aut	56.15	3487.95	No
12	9763-GRSh	Male	0	Yes	Yes	13	Yes	No	DSL	Yes	No	No	No	No	No	Month-to- Yes	Mailed check	49.95	587.45	No	
13	7469-LKBC	Male	0	No	No	16	Yes	No	No	No interne	No interne	No interne	No interne	No interne	No interne	Two year	No	Credit card (autorr	18.95	326.8	No
14	8091-TTVI	Male	0	Yes	No	58	Yes	Yes	Fiber optic	No	No	Yes	No	Yes	Yes	One year	No	Credit card (autorr	100.35	5681.1	No
15	0280-XIGE	Male	0	No	No	49	Yes	Yes	Fiber optic	No	Yes	Yes	No	Yes	Yes	Month-to- Yes	Bank transfer (aut	103.7	5036.3	Yes	
16	5129-JLPI	Male	0	No	No	25	Yes	No	Fiber optic	Yes	No	Yes	Yes	Yes	Yes	Month-to- Yes	Electronic check	105.5	2686.05	No	
17	3655-SNOQ	Female	0	Yes	Yes	69	Yes	Yes	Fiber optic	Yes	Yes	Yes	Yes	Yes	Yes	Two year	No	Credit card (autorr	113.25	7895.15	No
18	8191-XWS	Female	0	No	No	52	Yes	No	No	No interne	No interne	No interne	No interne	No interne	No interne	One year	No	Mailed check	20.65	1022.95	No
19	9959-WOF	Female	0	No	Yes	71	Yes	Yes	Fiber optic	Yes	No	Yes	No	Yes	Yes	Two year	No	Bank transfer (aut	106.7	7382.25	No
20	4190-MFLI	Female	0	Yes	Yes	10	Yes	No	DSL	No	No	Yes	Yes	No	No	Month-to- No	No	Credit card (autorr	55.2	528.35	Yes
21	4183-MYFI	Female	0	No	No	21	Yes	No	Fiber optic	No	Yes	No	No	No	Yes	Month-to- Yes	Electronic check	90.05	1862.9	No	
22	8779-QRD	Male	1	No	No	1	No	No phone	DSL	No	No	Yes	No	No	Yes	Month-to- Yes	Electronic check	39.65	39.65	Yes	
23	1680-VDCI	Male	0	Yes	No	12	Yes	No	No	No interne	No interne	No interne	No interne	No interne	No interne	One year	No	Bank transfer (aut	19.8	202.25	No
24	1066-JKSG	Male	0	No	No	1	Yes	No	No	No interne	No interne	No interne	No interne	No interne	No interne	Month-to- No	No	Mailed check	20.15	20.15	Yes
25	3638-WEA	Female	0	Yes	No	58	Yes	Yes	DSL	No	Yes	No	Yes	No	No	Two year	Yes	Credit card (autorr	59.9	3505.1	No

Important Columns

- customerID
- gender
- SeniorCitizen
- Partner
- Dependents
- tenure
- PhoneService
- MultipleLines
- InternetService
- OnlineSecurity
- OnlineBackup
- DeviceProtection
- TechSupport
- StreamingTV
- StreamingMovies
- Contract
- PaperlessBilling
- PaymentMethod
- MonthlyCharges
- TotalCharges
- Churn

5. Technology Stack

Category	Technology
Programming Language	Python
Data Processing	Pandas, NumPy
Machine Learning	Scikit-learn
Model Persistence	Joblib

Category	Technology
Database	PostgreSQL
ORM	SQLAlchemy
API Framework	FastAPI
API Server	Uvicorn
Dashboard	Metabase
Version Control	Git & GitHub

6. System Architecture

IBM Telco Dataset

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Data Cleaning & Preprocessing

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Feature Engineering

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Machine Learning Models

(Logistic Regression + LTV Regression)

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Predictions

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|—————▶ PostgreSQL Database

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|—————▶ FastAPI REST API

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|—————▶ Metabase Dashboard

7. Data Preprocessing

Data preprocessing is a critical step in any machine learning project. Raw datasets often contain missing values, inconsistent formats, duplicate records, and irrelevant information that can negatively affect model performance. Before training the machine learning models, the IBM Telco Customer Churn dataset was cleaned and transformed into a suitable format.

7.1 Loading the Dataset

The dataset was imported into Python using the Pandas library. It contains customer demographic information, subscribed services, billing details, and churn status.

```
import pandas as pd
```

```
df = pd.read_csv("telco.csv")
```

After loading the dataset, the first few rows and summary statistics were examined to understand the structure of the data.

7.2 Dataset Inspection

The following checks were performed:

- Number of rows and columns
- Data types
- Missing values
- Duplicate records
- Target variable distribution

The dataset contains approximately **7,043 customer records** with **21 columns**, including the target variable (**Churn**).

7.3 Handling Missing Values

The TotalCharges column contained blank values because some customers had recently joined the service and had no accumulated charges.

The following preprocessing steps were applied:

- Blank values were converted to NaN
- The column was converted from string to numeric
- Missing values were removed

```
df["TotalCharges"] = pd.to_numeric(df["TotalCharges"], errors="coerce")
```

```
df = df.dropna()
```

This ensured that all numerical calculations were performed correctly.

7.4 Data Type Conversion

Several columns required conversion into appropriate data types.

Examples include:

Column	Original Type	Converted Type
TotalCharges	Object	Float
SeniorCitizen	Integer	Integer
MonthlyCharges	Float	Float
tenure	Integer	Integer

Proper data types improve computational efficiency and model compatibility.

7.5 Encoding Categorical Variables

Machine learning algorithms cannot directly process categorical text values.

Therefore, categorical variables such as:

- Gender
- InternetService
- Contract
- PaymentMethod
- Partner
- Dependents

were converted into numerical values using encoding techniques.

For binary features, Label Encoding was applied.

For multi-class features, One-Hot Encoding was used through Scikit-learn pipelines.

7.6 Feature Scaling

Certain machine learning algorithms perform better when numerical features are on similar scales.

Continuous variables such as:

- MonthlyCharges
- TotalCharges
- Tenure

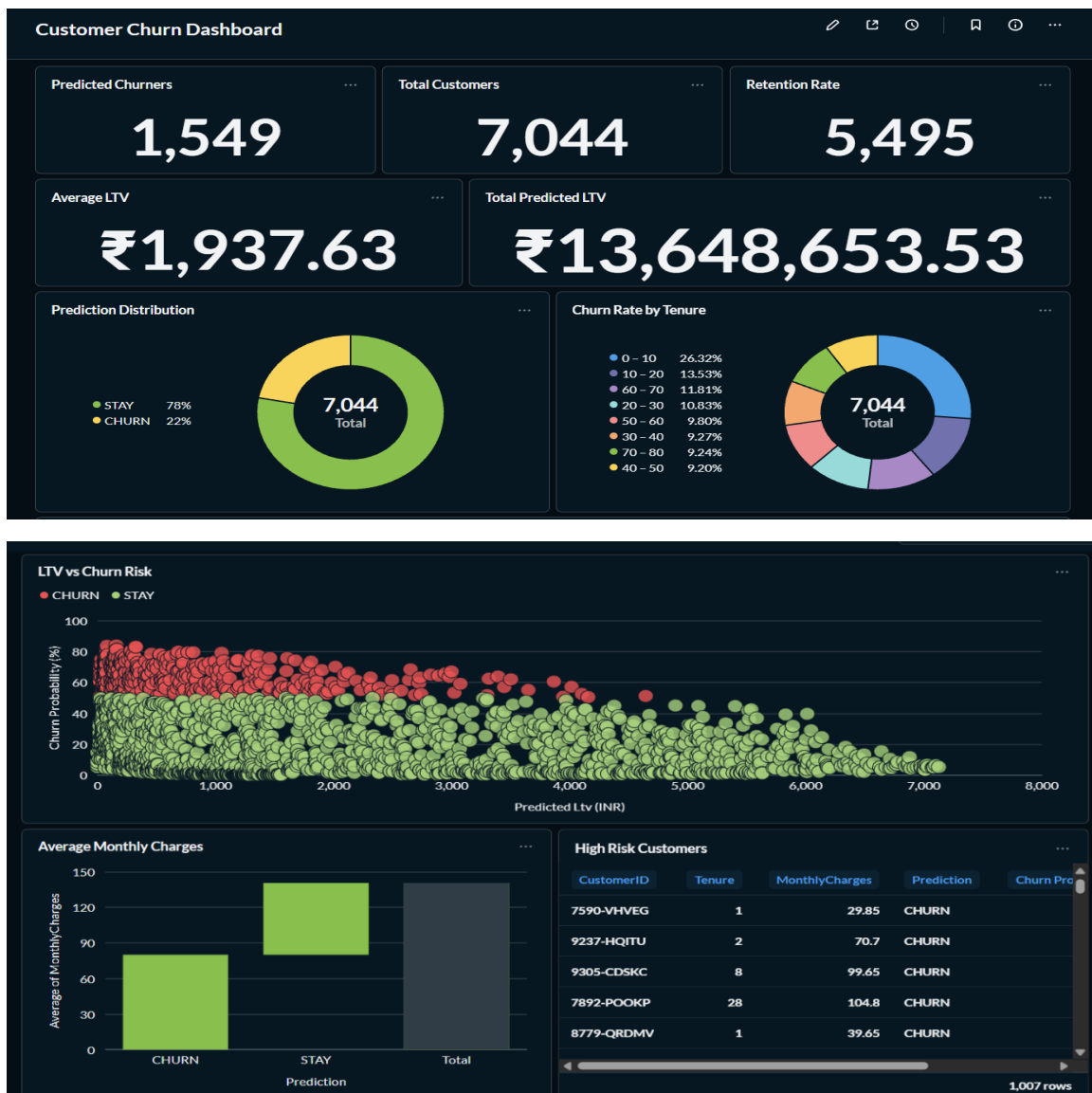
were standardized using **StandardScaler**.

Scaling improves convergence and helps Logistic Regression learn more effectively.

8. Exploratory Data Analysis (EDA)

Exploratory Data Analysis was conducted to understand customer behavior and identify important relationships within the dataset.

EDA helps reveal hidden trends that influence customer churn.



8.1 Churn Distribution

The first analysis examined the number of customers who stayed versus those who left.

Observation:

- Majority of customers remained with the company.
- Approximately one-quarter of customers had churned.

This indicates an imbalanced classification problem.

8.2 Contract Type Analysis

Customers were grouped according to contract type.

Findings:

- Month-to-month contracts showed the highest churn.
- One-year contracts had moderate churn.
- Two-year contracts had the lowest churn.

Interpretation:

Customers with longer commitments are less likely to leave the company.

8.3 Monthly Charges Analysis

Monthly charges were analyzed against churn.

Observation:

Customers paying higher monthly charges exhibited a higher probability of leaving.

Possible reasons include:

- High pricing
 - Better competitor offers
 - Low perceived service value
-

8.4 Tenure Analysis

Customer tenure represents the number of months a customer has remained with the company.

Observation:

Customers with shorter tenure churned more frequently than long-term customers.

Interpretation:

Retaining customers during the initial months is essential.

8.5 Internet Service Analysis

The dataset contains different internet service types:

- DSL
- Fiber Optic
- No Internet Service

Observation:

Customers using Fiber Optic services showed comparatively higher churn rates.

Possible reasons include:

- Higher pricing
 - Service quality expectations
 - Competitive alternatives
-

8.6 Payment Method Analysis

Different payment methods were compared.

Customers using electronic check payments exhibited higher churn compared to customers using automatic bank transfers or credit cards.

8.7 Correlation Analysis

A correlation matrix was generated for numerical variables.

Important observations:

- TotalCharges strongly correlates with tenure.
 - MonthlyCharges positively influences TotalCharges.
 - Tenure negatively correlates with churn.
-

9. Feature Engineering

Feature engineering was performed to create additional business-oriented variables that improve prediction performance.

Instead of relying only on the original dataset, several new features were derived.

9.1 Revenue Per Month

A new feature was created to measure the average monthly revenue generated by each customer.

Formula:

$\text{Revenue_Per_Month} = \text{TotalCharges} / \text{Tenure}$

This helps distinguish customers with similar total charges but different subscription durations.

9.2 Charge Ratio

Charge Ratio compares current monthly charges with accumulated spending.

$\text{Charge_Ratio} = \text{MonthlyCharges} / \text{TotalCharges}$

This provides insight into customer payment behavior.

9.3 Support Services Count

Customers subscribing to additional support services generally show better retention.

A new feature counted the number of subscribed services:

- Online Security
- Online Backup
- Device Protection
- Tech Support

Higher values indicate stronger customer engagement.

9.4 Streaming Services Count

Streaming-related services were combined into another feature.

It counts:

- Streaming TV
- Streaming Movies

Customers actively using multiple services generally exhibit different churn patterns.

9.5 Benefits of Feature Engineering

The engineered features improved the model by:

- Capturing customer behavior more effectively
 - Reducing information loss
 - Improving prediction accuracy
 - Increasing business interpretability
 - Enhancing model robustness
-

10. Data Splitting

After preprocessing and feature engineering, the dataset was divided into training and testing sets.

Typical split:

- **Training Data:** 80%
- **Testing Data:** 20%

The training dataset was used to build the machine learning models, while the testing dataset evaluated model performance on unseen data.

This separation helps ensure that the models generalize well and do not simply memorize the training data.

11. Machine Learning Models

Machine learning is the core component of the Customer Churn Prediction & Lifetime Value (LTV) Engine. After preprocessing and feature engineering, predictive models were trained to solve two business problems:

1. Predict whether a customer is likely to churn (Classification).
2. Estimate the future lifetime value (LTV) of each customer (Regression).

The dataset was divided into training and testing sets using an 80:20 ratio. The training set was used to build the models, while the testing set evaluated their performance on unseen data.

11.1 Churn Prediction Model

Customer churn prediction is a **binary classification problem**, where the target variable (**Churn**) has two possible values:

- **Yes** – Customer is likely to leave the company.
- **No** – Customer is expected to remain with the company.

Several algorithms can solve this problem, including Decision Trees, Support Vector Machines, Random Forests, and Logistic Regression. For this project, **Logistic Regression**

was selected because it provides strong performance, is computationally efficient, and produces probability scores that are useful for business decisions.

Why Logistic Regression?

The following reasons influenced the choice of Logistic Regression:

- Performs well for binary classification problems.
- Easy to interpret.
- Fast training and prediction.
- Produces probability scores for churn risk.
- Suitable for structured tabular data.

11.2 Logistic Regression Workflow

The workflow used to build the churn prediction model is shown below:

Preprocessed Dataset

|



Feature Selection

|



Train-Test Split

|



Logistic Regression

|



Prediction

|



Performance Evaluation

11.3 Model Training

The Logistic Regression model was trained using Scikit-learn after applying preprocessing and feature engineering.

The training process involved:

- Selecting input features
- Splitting data into training and testing sets
- Standardizing numerical variables
- Training the classifier
- Predicting customer churn
- Evaluating model performance

11.4 Model Performance

The trained model achieved an overall classification accuracy of approximately **82%** on the testing dataset.

Performance metrics used for evaluation include:

- Accuracy
- Precision
- Recall
- F1-Score
- Confusion Matrix

Each metric provides a different perspective on model performance.

Accuracy

Accuracy measures the percentage of correctly classified customers.

Formula

Accuracy =

Correct Predictions

Total Predictions

Higher accuracy indicates better overall performance.

Precision

Precision measures how many customers predicted as churn actually churned.

High precision reduces false alarms.

Recall

Recall measures how many actual churn customers were correctly identified.

A high recall is important because missing a customer who is about to leave may result in revenue loss.

F1-Score

The F1-score combines precision and recall into a single metric.

It provides a balanced measure of model performance.

Confusion Matrix

The confusion matrix summarizes prediction outcomes.

Actual / Predicted Churn		No Churn
Churn	True Positive	False Negative
No Churn	False Positive	True Negative

This helps evaluate classification errors.

12. Lifetime Value (LTV) Prediction

Customer Lifetime Value estimates the future revenue expected from each customer throughout their relationship with the company.

Unlike churn prediction, LTV estimation is a **regression problem** because the output is a continuous numerical value.

12.1 Why Predict LTV?

Predicting lifetime value enables businesses to:

- Identify premium customers.
- Allocate marketing budgets efficiently.
- Prioritize customer retention.
- Increase long-term profitability.

- Improve personalized marketing strategies.

12.2 Regression Model

A regression model was trained to estimate customer lifetime value using customer demographics, billing information, service subscriptions, and engineered features.

The model predicts the expected revenue contribution of each customer.

12.3 Evaluation Metrics

The LTV model was evaluated using:

Mean Absolute Error (MAE)

MAE measures the average difference between predicted and actual lifetime value.

A lower MAE indicates better prediction accuracy.

R² Score

The coefficient of determination (R²) measures how well the regression model explains variation in customer lifetime value.

Higher values indicate stronger predictive performance.

12.4 LTV Prediction Workflow

Customer Features

|



Regression Model

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Predicted Lifetime Value

|



Business Decision

13. Feature Importance

Understanding the most influential features helps businesses identify the primary reasons behind customer churn and high customer value.

The most significant features identified during model analysis include:

Feature	Business Importance
Contract	Long-term contracts reduce churn.
Tenure	Long-standing customers are less likely to leave.
MonthlyCharges	High monthly bills increase churn risk.
TotalCharges	Indicates overall customer spending.
TechSupport	Customers with support services are more likely to stay.
OnlineSecurity	Security services improve customer retention.
Revenue_Per_Month	Indicates monthly revenue generated by the customer.
Support_Services	Higher engagement reduces churn probability.

These features provide valuable business insights beyond predictive accuracy.



14. Business Interpretation of Results

The machine learning models generated several actionable insights:

Long-Term Contracts

Customers with one-year and two-year contracts exhibited significantly lower churn rates than those on month-to-month contracts.

Business Recommendation: Offer incentives for customers to switch to long-term contracts.



High Monthly Charges

Customers paying higher monthly fees showed a greater tendency to churn.

Business Recommendation: Introduce customized discounts or loyalty rewards for high-paying customers.



Customer Tenure

Customers with short tenure were more likely to leave.

Business Recommendation: Focus retention campaigns on customers during their first six months of service.



Support Services

Customers using services such as Tech Support, Online Backup, and Online Security demonstrated lower churn.

Business Recommendation: Encourage adoption of value-added services through bundled packages.

High LTV Customers

Customers with high predicted lifetime value should receive priority retention efforts because losing them would have a greater financial impact.

Business Recommendation: Develop personalized retention programs for premium customers.

15. Summary

The machine learning component successfully achieved the project's objectives by:

- Predicting customer churn with strong classification performance.
- Estimating customer lifetime value.
- Identifying important factors influencing customer retention.
- Providing actionable insights for business decision-making.
- Supporting proactive customer retention strategies.

The combination of churn prediction and LTV estimation creates a comprehensive analytics solution that helps organizations reduce customer loss and maximize long-term revenue.

16. Database Design

16.1 Overview

To efficiently manage customer information and prediction results, PostgreSQL was used as the backend relational database. PostgreSQL provides reliability, scalability, and excellent support for analytical queries.

The database stores both the original customer data and the predictions generated by the machine learning models.

16.2 Database Schema

The project consists of two primary tables:

Table 1: customers

Column	Data Type	Description
customerID	VARCHAR	Unique customer identifier
gender	VARCHAR	Customer gender
SeniorCitizen	INTEGER	Senior citizen status
Partner	VARCHAR	Whether customer has a partner
Dependents	VARCHAR	Whether customer has dependents
tenure	INTEGER	Number of months with company
Contract	VARCHAR	Contract type
MonthlyCharges	FLOAT	Monthly bill amount
TotalCharges	FLOAT	Total amount charged
Churn	VARCHAR	Actual churn status

Table 2: customer_predictions

Column	Data Type	Description
customerID	VARCHAR	Customer ID
ChurnPrediction	VARCHAR	Predicted churn
ChurnProbability	FLOAT	Probability of churn
PredictedLTV	FLOAT	Estimated lifetime value

16.3 Why PostgreSQL?

The following reasons motivated the use of PostgreSQL:

- Open-source and free
- High performance
- ACID-compliant transactions
- Excellent integration with SQLAlchemy
- Suitable for analytical workloads

- Easily connects with Metabase for visualization

17. FastAPI Implementation

17.1 Overview

FastAPI was used to expose the trained machine learning models through REST APIs.

This allows other applications to request predictions in real time without retraining the models.

The screenshot displays the FastAPI Swagger UI. At the top, the 'FastAPI' logo is shown with version '0.1.0' and 'OAS 3.1' badges. Below the logo is a link to '/openapi.json'. The main section is titled 'default' and lists two endpoints: a GET endpoint at '/' labeled 'Home' and a POST endpoint at '/predict_churn' labeled 'Predict'. The POST endpoint is selected, and its response is shown in a light green panel. The response status is '200'. The 'Response body' is a JSON object:

```
{  "prediction": "CHURN",  "probability": 73.08}
```

. The 'Response headers' are:

```
content-length: 42
content-type: application/json
date: Fri, 19 Jun 2026 13:10:09 GMT
server: uvicorn
```

. At the bottom, there is a 'Responses' section.

FastAPI 0.1.0 OAS 3.1
</openapi.json>

default

GET / Home

POST /predict_churn Predict

Code **Details**

200

Response body

```
{  "prediction": "CHURN",  "probability": 73.08}
```

Response headers

```
content-length: 42
content-type: application/json
date: Fri, 19 Jun 2026 13:10:09 GMT
server: uvicorn
```

Responses

17.2 API Workflow

Client Request

|



FastAPI Endpoint

|



Load Trained Model

|



Generate Prediction

|



Return JSON Response

17.3 API Endpoints

Health Check

GET /

Purpose:

Checks whether the API server is running.

Example Response

```
{  
  "status": "API Running Successfully"  
}
```

Predict Customer Churn

POST /predict

Input

Customer information in JSON format.

Output

```
{  
  "prediction": "No",  
  "probability": 0.87  
}
```

Predict Customer Lifetime Value

POST /predict_ltv

Output

```
{  
  "Predicted_LTV": 12543.87  
}
```

17.4 Advantages of FastAPI

- High performance
 - Automatic API documentation
 - Easy deployment
 - JSON-based communication
 - Scalable architecture
-

18. Metabase Dashboard

18.1 Overview

Metabase was used to create interactive dashboards for business users.

Instead of manually writing SQL queries, decision-makers can view insights through visual charts and KPIs.

18.2 Dashboard Components

The dashboard contains:

KPI Cards

- Total Customers
- Active Customers
- Churned Customers

- Churn Rate
 - Average Predicted LTV
-

Charts

- Churn by Contract Type
 - Churn by Internet Service
 - Monthly Charges Distribution
 - Customer Tenure Analysis
 - Revenue Distribution
 - High-Value Customers
-

18.3 Business Benefits

The dashboard enables managers to:

- Monitor customer retention
 - Identify high-risk customers
 - Analyze revenue trends
 - Track churn patterns
 - Support strategic decision-making
-

19. Project Folder Structure

Customer-Churn-LTV-Engine/

```
|
|
|— data/
|  |— raw/
|  |— processed/
|
|— notebooks/
|
|— src/
|  |— ingestion/
```

```
| |— preprocessing/
| |— models/
| |— feature_engineering/
| |— database/
|
|— api/
| |— main.py
|
|— dashboard/
|
|— models/
| |— churn_pipeline.pkl
| |— ltv_model.pkl
|
|— requirements.txt
|— README.md
|— .gitignore
```

20. Installation Guide

Step 1

Clone the repository.

```
git clone https://github.com/yourusername/Customer-Churn-LTV-Engine.git
```

Step 2

Create a virtual environment.

```
python -m venv venv
```

Step 3

Activate the virtual environment.

Windows


```
venv\Scripts\activate
```

Linux/Mac

```
source venv/bin/activate
```

Step 4

Install dependencies.

```
pip install -r requirements.txt
```

Step 5

Configure PostgreSQL.

- Create the database.
 - Update database credentials.
 - Load the dataset.
-

Step 6

Run the FastAPI server.

```
uvicorn api.main:app --reload
```

Step 7

Launch Metabase.

Connect PostgreSQL and import the dashboards.

21. Results & Business Insights

The developed system successfully predicts customer churn and estimates customer lifetime value using historical telecom customer data.

Key Results

- Churn prediction accuracy: **approximately 82%**
- End-to-end ML pipeline developed
- Prediction results stored in PostgreSQL
- REST API successfully deployed using FastAPI
- Interactive dashboard created using Metabase

Business Insights

- Customers with month-to-month contracts have the highest churn risk.
 - Short-tenure customers are more likely to leave.
 - Higher monthly charges are associated with increased churn.
 - Customers using support services exhibit lower churn.
 - High-LTV customers should receive personalized retention strategies.
-

22. Challenges Faced

During development, several technical challenges were encountered:

- Handling missing values in the TotalCharges column.
- Encoding multiple categorical variables.
- Feature engineering for improved prediction.
- Configuring PostgreSQL connectivity.
- Managing SQLAlchemy connection strings.
- Integrating FastAPI with trained machine learning models.
- Creating SQL queries for Metabase dashboards.

These challenges were resolved through iterative testing, debugging, and validation.

23. Future Enhancements

The project can be further improved by implementing:

- Deep Learning models for enhanced prediction accuracy.
 - SHAP-based explainable AI visualizations.
 - Real-time streaming predictions using Apache Kafka.
 - Automated model retraining pipelines.
 - Docker containerization.
 - Cloud deployment on AWS, Azure, or Google Cloud.
 - Role-based web application for business users.
 - Automated email alerts for high-risk customers.
-

24. Conclusion

The **Customer Churn Prediction & Lifetime Value (LTV) Engine** demonstrates how machine learning can be applied to solve real business challenges in the telecommunications industry. By combining churn prediction and lifetime value estimation into a unified analytics platform, the project enables organizations to proactively identify customers at risk of leaving while prioritizing retention efforts based on customer value.

The implementation integrates data preprocessing, feature engineering, predictive modeling, PostgreSQL database management, FastAPI-based REST APIs, and Metabase dashboards into a complete end-to-end workflow. This architecture provides accurate predictions, supports informed business decisions, and offers a scalable foundation for future enhancements.

Overall, the project highlights the practical use of data analytics and machine learning in improving customer retention, optimizing marketing strategies, and maximizing long-term business revenue.

25. References

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